

EDS ASSIGNMENT

NAME: VARSHARANI SHIVAJI PATIL

DIVISION:SE2

BATCH:S22

PRN:202501100152

DATASET: WordNet Dataset

LINK TO DATASET:

<https://www.kaggle.com/datasets/duketemon/wordnet-synonyms>

```
df = pd.read_csv("wordnet.csv")
```

IMPORTANT DATASET FIELDS:

- word
- synset_id
- definition
- pos
- frequency
- hypernym

1. Find most common part of speech

Problem: Which part of speech (noun, verb, etc.) is most frequent?

ANS:

```
df['pos'].value_counts().idxmax()
```

2. Find word with highest frequency

Problem: Which word appears most frequently?

ANS:

```
df[df['frequency'] == df['frequency'].max()]
```

3. Count total number of words

Problem: Total number of words present in dataset?

ANS:

```
len(df)
```

4. Find words with low frequency

Problem: Show words with frequency less than 5.

ANS:

```
df[df['frequency'] < 5]
```

5. Count words with high frequency

Problem: How many words have frequency greater than 50?

ANS:

```
df[df['frequency'] > 50].shape[0]
```

6. Find average frequency

Problem: What is the average frequency of words?

ANS:

```
df['frequency'].mean()
```

7. Find longest word

Problem: Which word has maximum length?

ANS:

```
df['word_length'] = df['word'].apply(len)
df[df['word_length'] == df['word_length'].max()]
```

8. Find shortest word

Problem: Which word has minimum length?

ANS:

```
df[df['word_length'] == df['word_length'].min()]
```

9. Count number of synsets

Problem: Total number of unique synsets in dataset?

ANS:

```
df['synset_id'].nunique()
```

10. Find words without hypernyms

Problem: Show words that do not have a hypernym.

ANS:

```
df[df['hypernym'].isnull()]
```

11. Find top 5 most frequent words

Problem: Show top 5 most commonly used words.

ANS:

```
df.sort_values(by='frequency', ascending=False).head(5)
```

12. Group words by part of speech

Problem: Count number of words in each part of speech.

ANS:

```
df['pos'].value_counts()
```

13. Find words starting with vowel

Problem: Show words that start with vowels.

ANS:

```
df[df['word'].str.startswith(('a','e','i','o','u'))]
```

14. Find shortest definition

Problem: Which word has the smallest definition length?

ANS:

```
df['def_len'] = df['definition'].apply(len)
```

```
df[df['def_len'] == df['def_len'].min()]
```

15. Count missing values

Problem: How many missing values are present in dataset?

ANS:

```
df.isnull().sum()
```

16. Replace missing hypernyms

Problem: Replace missing hypernyms with "Unknown".

ANS:

```
df['hypernym'].fillna("Unknown", inplace=True)
```

17. Normalize frequency using NumPy

Problem: Normalize frequency values between 0 and 1.

ANS:

```
import numpy as np
```

```
df['norm_freq'] = (df['frequency'] - np.min(df['frequency'])) /  
(np.max(df['frequency']) - np.min(df['frequency']))
```

18. Find median frequency

Problem: What is median frequency of words?

ANS:

```
import numpy as np
```

```
np.median(df['frequency'])
```

19. Find duplicate words

Problem: Show duplicate words in dataset.

ANS:

```
df[df.duplicated('word')]
```

20. Check relation between word length and frequency

Problem: Is there any relationship between word length and frequency?

ANS:

```
import numpy as np
```

```
np.corrcoef(df['word_length'], df['frequency'])
```